

Table 1

Command	Action	Description
AT<CR>	Checks the connection between the computer and the GSM device	The modem responds to this command with <i>OK</i> or <i>ERROR</i> . <i>OK</i> indicates the modem is ready.
AT+CPIN?<CR>	Checks the SIM in the device	This command is meant for password query, but can be used to check the SIM. If the response is <i>READY</i> , it means that the SIM is okay.
AT+CMGF=1<CR>	Sets the SMS message format	This command sets the message format to text mode.
AT+CSCA=<address><CR>	Sets the SMS service centre address	In this command, the keyword <i>address</i> represents the SMS service centre address to be set.
AT+CSQ<CR>	Checks signal quality	This command indicates the strength of the received signal.
AT+CMGS=<number><CR><message><ctrl-z>	Sends an SMS to the GSM network	This command sends the message to the network. Here, the number and message represent the recipient mobile number and the SMS text message string respectively. To indicate the end of the message body, <ctrl-z> must be used.
AT+CMGL=<stat><CR>	Reads messages from the inbox	'Stat' can be <i>REC UNREAD</i> , <i>REC READ</i> , <i>STO UNSENT</i> , <i>STO SENT</i> , and <i>ALL</i> for received unread messages, received read messages, stored unsent messages, stored sent messages and all messages, respectively. This command returns a message with the status value from the preferred message storage location.
AT+CMGD=<index><CR>	Deletes SMSs from the inbox	This command deletes the message from the preferred message storage location. 'Index' represents the location number of the SMS to be deleted.

to interconnect the machines. With this approach, the server executing on the gateway machine will continuously be listening to the clients. Once a client connects to the server, the client can send the mobile number and SMS text to the server for transfer. After extracting the information, the server prepares the appropriate AT commands and sends the SMS to the GSM network. The software can further be extended to send SMS alerts. Mobile numbers of those who receive alerts can be registered in the database and if the desired event can be fed to gateway software, the software can create the corresponding response and the message can be sent as an SMS alert to the registered mobile number.

Summary

The AT commands can be implemented in software to

exchange SMSs between a PC and the GSM network. The concept can further be extended to generate SMS alerts for important events. 

References

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